

Probability and Sampling

Distributions, weighted dice, and randomness · video 1:45

This lesson is about what an LLM actually outputs: not a word, but a **probability distribution** over words. You will see what a probability is, why a distribution adds up to one, how **sampling** picks one word at random according to it (a weighted spinner), and how a **seed** makes that randomness repeatable.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 10).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. A probability is a number from 0 to 1	▶ 0:08-0:26	4	☐
2. A distribution adds up to one	▶ 0:26-0:43	4	☐
3. Sampling: a weighted spinner	▶ 0:43-0:52	4	☐
4. Many spins match the probabilities	▶ 0:53-1:02	4	☐
5. Why sample? Variety	▶ 1:02-1:10	4	☐
6. Seeds make randomness repeatable	▶ 1:10-1:22	3	☐
7. Sampling in NumPy	▶ 1:23-1:43	3	☐

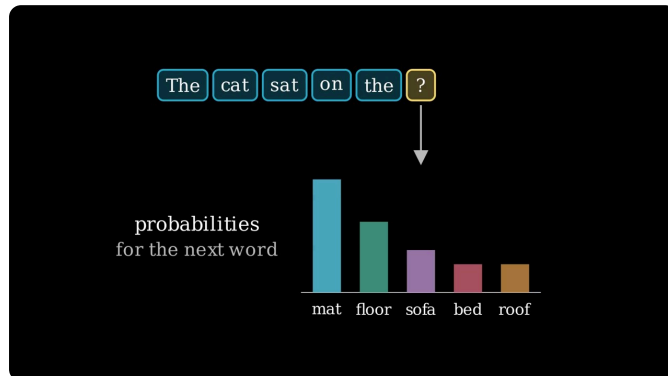
Before you start: no earlier Foundations video is needed. The whole lesson uses the video's illustrative distribution for the word after “The cat sat on the”: *mat* 0.40, *floor* 0.25, *sofa* 0.15, *bed* 0.10, *roof* 0.10. You will see these ideas in episode 1 (the model predicts a distribution), episode 10 (sampling from it) and episode 14.

CONCEPT 1 OF 7

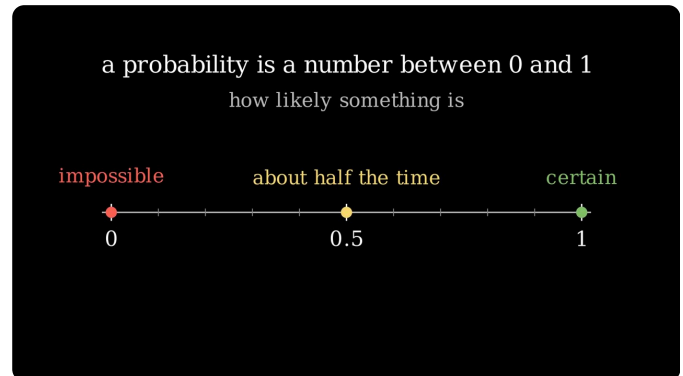
A probability is a number from 0 to 1

▶ Video 0:08–0:26

An LLM never simply knows the next word. It produces **probabilities**. A probability is a number **between 0 and 1** that says how likely something is: **0** means impossible, **1** means certain, and **0.5** means it happens about half the time. As a percentage, 0.25 is 25 %.



▶ 0:14 An LLM never simply knows the next word: it produces a probability for each candidate.



▶ 0:25 0 means impossible, 1 means certain, 0.5 means about half the time.

KEY IDEA

$0 \leq p \leq 1$. The bigger p , the more often the event happens: $p = 0.5$ about half the time, $p = 0.1$ about one time in ten.

Check 1: A probability is a number from 0 to 1

Answer all 4 questions before looking at the answers. Pass mark: all correct.

1.1 MULTIPLE CHOICE

Which of these can be a probability?

- A. 1.2
- B. -0.1
- C. 0.35
- D. $3/2$

1.2 CALCULATION

An event has probability 0.1. In 1,000 tries, about how many times does it happen? And an event with probability 0.5?

1.3 TRUE OR FALSE

"An event with probability 0.001 is impossible."

True ☐ False ☐

1.4 PUT IN ORDER

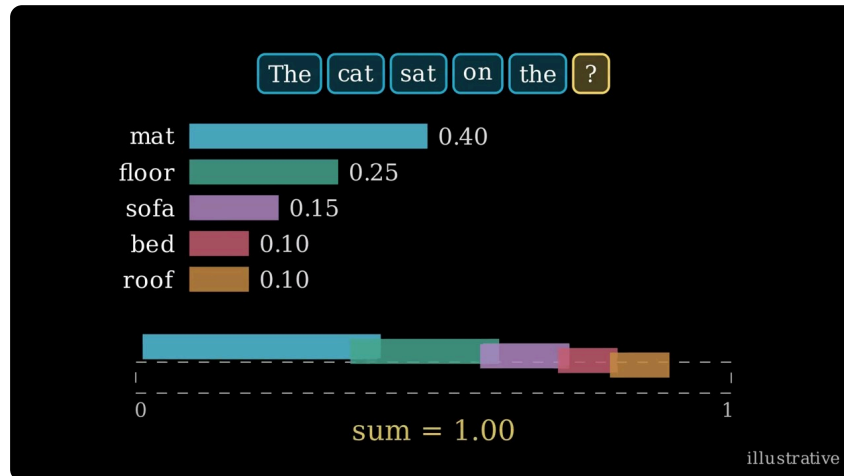
Put in order from least to most likely: $p = 0.5$ · certain · $p = 0.05$ · impossible · $p = 0.9$.

CONCEPT 2 OF 7

A distribution adds up to one

▶ Video 0:26–0:43

A **probability distribution** spreads **one whole unit of belief** across all the options. For the word after “*The cat sat on the*”, the video's illustrative distribution is *mat* 0.40, *floor* 0.25, *sofa* 0.15, *bed* 0.10 and *roof* 0.10: together **exactly 1**. So the chance of “bed or roof” is the sum of their shares, and “not mat” gets whatever is left, $1 - 0.40 = 0.60$.



▶ 0:42 One whole unit, shared out across the options: sum = 1.00.

KEY IDEA

Every value is between 0 and 1, and together they **add up to exactly 1**. More for one option means less for the others.

Check 2: A distribution adds up to one

Answer all 4 questions before looking at the answers. Pass mark: all correct.

2.1 CALCULATION

Using the video's distribution: (a) check that it adds up to 1. What is the probability that the next word is (b) not *mat*, (c) *bed* or *roof*?

2.2 MULTIPLE CHOICE

Which could be a distribution over the words *yes*, *no*, *maybe*?

- A. 0.6, 0.3, 0.3
- B. 0.8, 0.2, 0.0
- C. 0.9, 0.2, -0.1
- D. 0.3, 0.3, 0.3

2.3 TRUE OR FALSE

“A model raises *mat* from 0.40 to 0.60. The other four words can keep exactly the same probabilities.”

True ☐ False ☐

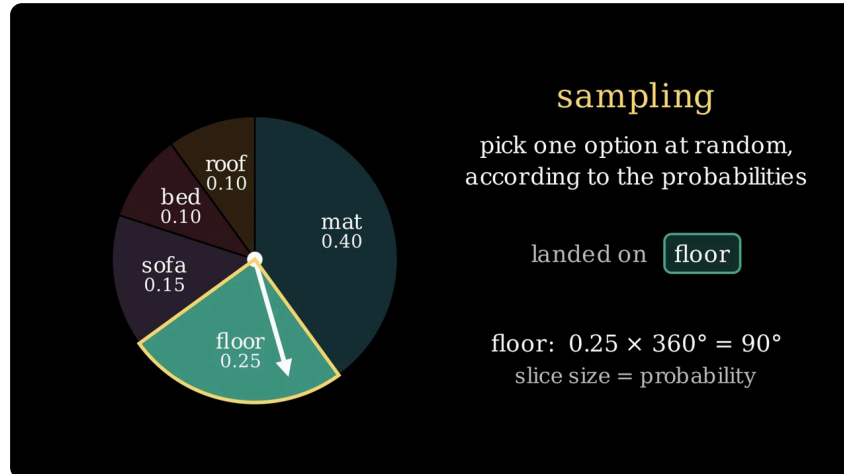
2.4 CALCULATION

For the word after “*I drank a cup of*”, a model gives *tea* 0.55, *coffee* 0.30 and *water* 0.10. How much probability is left for all the other words together?

Sampling: a weighted spinner

▶ Video 0:43–0:52

Sampling means picking one option **at random, according to the probabilities**. Think of a spinner where each word gets a **slice sized by its probability**: *floor* gets 0.25 of the circle, $0.25 \times 360^\circ = 90^\circ$. Spin it and it lands on one word. This time it landed on *floor*, not on the favourite, *mat*.



▶ 0:52 A spinner with one slice per word. floor's slice is $0.25 \times 360^\circ = 90^\circ$; this spin landed on floor.

KEY IDEA

Slice size = probability. One sample = one spin: a random pick in which every option can win, each with its own probability.

Check 3: Sampling: a weighted spinner

Answer all 4 questions before looking at the answers. Pass mark: all correct.

3.1 CALCULATION

Work out the slice angle of every word: *mat* 0.40, *floor* 0.25, *sofa* 0.15, *bed* 0.10, *roof* 0.10. Do they fill the circle?

3.2 CALCULATION

On another spinner, a slice covers 72° . What probability does it stand for?

3.3 MULTIPLE CHOICE

The spinner landed on *floor*. What does that tell you?

- A. floor is the most likely word
- B. This spin happened to land on floor, which has a 25 % chance on every spin
- C. The next spin will also give floor
- D. mat's probability went down

3.4 TRUE OR FALSE

"Because *mat* has the biggest slice, sampling always picks *mat*."

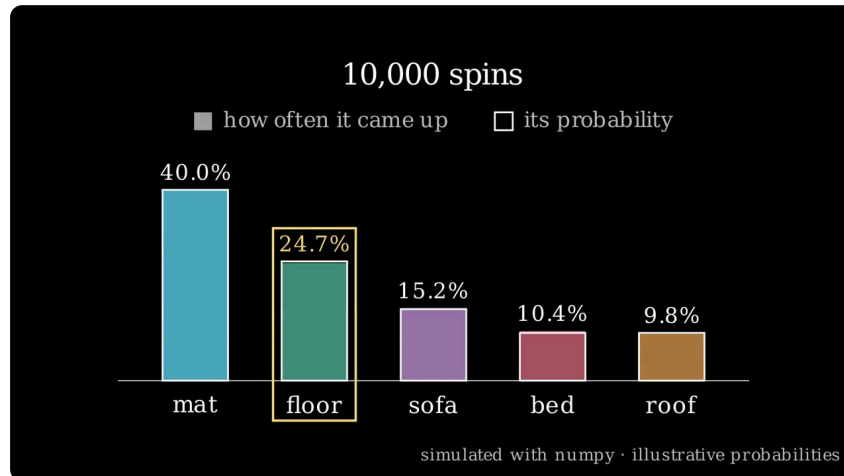
True ☐ False ☐

CONCEPT 4 OF 7

Many spins match the probabilities

▶ Video 0:53–1:02

One spin is unpredictable, but many spins are not. Spin 10,000 times and each word comes up **about as often as its probability says**: in the video's simulation, *mat* 40.0 %, *floor* 24.7 %, *sofa* 15.2 %, *bed* 10.4 % and *roof* 9.8 %. The match is close but not exact, and it gets better the more times you spin. With only a few spins, the counts can be far off.



▶ 1:01 10,000 simulated spins: how often each word came up (filled) against its probability (outline).

KEY IDEA

Over many samples, **how often $\approx$ probability**. Expected count = probability $\times$ number of samples.

Check 4: Many spins match the probabilities

Answer all 4 questions before looking at the answers. Pass mark: all correct.

4.1 CALCULATION

You sample 200 times from the video's distribution. About how many times do you expect each word?

.....

4.2 CALCULATION

In the 10,000 spins, *floor* came up 24.7 % of the time. About how many times is that? How many would its probability predict?

.....

4.3 SHORT ANSWER

Ten samples gave: *bed, sofa, mat, mat, bed, mat, bed, sofa, floor, mat*. Count each word. Which word is furthest from its probability? Is the sampler broken?

.....

4.4 TRUE OR FALSE

"After 10,000 spins, every word came up exactly as often as its probability says."

True ☐ False ☐

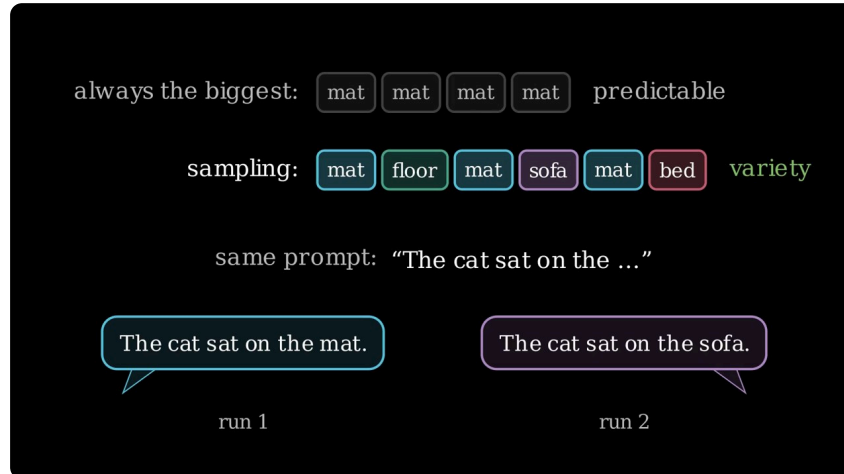
CONCEPT 5 OF 7

Why sample? Variety

▶ Video 1:02-1:10

Always choosing the biggest slice (**greedy** picking) is **predictable**: *mat, mat, mat*, every time.

Sampling adds **variety**. That is why the same prompt can give different answers: one run says “*The cat sat on the mat.*”, the next “*The cat sat on the sofa.*”



▶ 1:09 Always taking the biggest slice repeats itself; sampling varies, so the same prompt can give different answers.

KEY IDEA

Greedy = always the most likely option: the same answer every time. Sampling = a random pick by probability: varied answers, with likely words still the most common.

Check 5: Why sample? Variety

Answer all 4 questions before looking at the answers. Pass mark: all correct.

5.1 CALCULATION

With the video's distribution, what is the probability that one sample is **not** the greedy choice?

.....

5.2 CALCULATION

You run the prompt 20 times, sampling the next word each time. About how many runs end in *sofa*? And with greedy picking?

.....

5.3 MULTIPLE CHOICE

Why can a chatbot give two different answers to the same prompt?

- A. The model's probabilities changed between the runs
- B. It samples from its probabilities
- C. It forgot the prompt
- D. Greedy picking is random

5.4 TRUE OR FALSE

“With greedy picking, the same prompt always gives the same next word.”

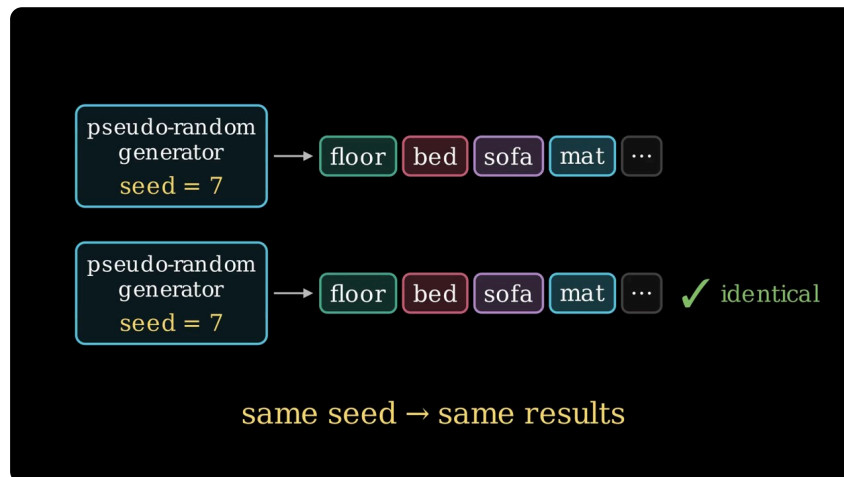
True ☐ False ☐

Seeds make randomness repeatable

▶ Video 1:10-1:22

Computers make randomness with **pseudo-random number generators**: a fixed recipe that produces numbers that look random. The recipe starts from a number called the **seed**. Start one with the **same seed** and you get the **same sequence** of random choices every time. With seed 7 and the video's distribution, it always begins *floor, bed, sofa, mat*.

That keeps experiments **reproducible**: you, or anyone else, can rerun the code and get exactly the same “random” results.



▶ 1:22 Two generators started with seed 7 give identical choices: floor, bed, sofa, mat, ...

KEY IDEA

Same seed → same results. Pseudo-random choices look random, but the seed fixes the whole sequence.

Check 6: Seeds make randomness repeatable

Answer all 3 questions before looking at the answers. Pass mark: all correct.

6.1 TRUE OR FALSE

“Two generators started with the same seed can give different sequences.”

True ☐ False ☐

6.2 MULTIPLE CHOICE

Why do researchers fix the seed of their experiments?

- A. To make the model more accurate
- B. So the experiment can be repeated exactly
- C. To change the probabilities
- D. To make sampling faster

6.3 SHORT ANSWER

You sample with seed 7; a colleague uses seed 8. Must your first sampled words be the same? Over 10,000 samples, should your word frequencies be similar?

.....

.....

Sampling in NumPy

▶ Video 1:23-1:43

In NumPy, create a generator with a seed, then call `choice` with the options and their probabilities `p`. Without `size` it returns one word; with `size=10`, ten:

```

1 words = ["mat", "floor", "sofa", "bed", "roof"]
2 p = [0.4, 0.25, 0.15, 0.1, 0.1] # adds up to 1
3 rng = np.random.default_rng(seed=7)
4 rng.choice(words, p=p) # 'floor'
5 rng.choice(words, size=10, p=p) # ten samples

```

one: floor

ten: bed sofa mat mat bed mat bed sofa floor mat

▶ 1:29 The code from the video: a seeded generator samples one word, then ten.

```

words = ["mat", "floor", "sofa", "bed", "roof"]
p = [0.4, 0.25, 0.15, 0.1, 0.1] # adds up to 1
rng = np.random.default_rng(seed=7)
rng.choice(words, p=p) # 'floor'
rng.choice(words, size=10, p=p) # ten samples

```

You'll see this in episode 1, where the model predicts a distribution, in episode 10, where we sample from it, and in episode 14.

KEY IDEA

`rng = np.random.default_rng(seed)`, then `rng.choice(options, size=n, p=probs)`. Same seed and same calls → same samples.

Check 7: Sampling in NumPy

Answer all 3 questions before looking at the answers. Pass mark: all correct.

7.1 MULTIPLE CHOICE

What does `rng.choice(words, size=10, p=p)` return?

- A. The 10 most likely words
- B. Ten words, each picked at random according to `p`
- C. Ten copies of 'mat'
- D. The probabilities of ten words

7.2 TRUE OR FALSE

"With `p = [0.4, 0.25, 0.15, 0.1, 0.2]`, NumPy refuses to sample."

True ☐ False ☐

7.3 HANDS-ON CODE

Try it yourself. (a) Run the code. Do the first two printed lines match the video? (b) The third print starts seed 7 again but asks for ten words at once. **Predict** it before you run it, then explain. (c) What do the last five lines print, and which frame of the video do they match?

```
import numpy as np

words = ["mat", "floor", "sofa", "bed", "roof"]
p = [0.4, 0.25, 0.15, 0.1, 0.1]

rng = np.random.default_rng(seed=7)
print(rng.choice(words, p=p))          # one word
print(rng.choice(words, size=10, p=p))  # ten more

rng = np.random.default_rng(seed=7)    # start again
print(rng.choice(words, size=10, p=p))  # ten at once

s = np.random.default_rng(seed=0).choice(words, size=10000, p=p)
for w in words:
    print(w, (s == w).mean())
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. A probability is a number from 0 to 1 · rewatch ▶ 0:08-0:26

1.1 C. A probability must lie between 0 and 1. 1.2 and $3/2$ are above 1; -0.1 is below 0.

1.2 About 100; about 500. $0.1 \times 1,000 = 100$ and $0.5 \times 1,000 = 500$. “About”, because chance makes the exact count vary.

1.3 False. It is very unlikely (about once in 1,000 tries), but possible. Only 0 means impossible.

1.4 impossible → 0.05 → 0.5 → 0.9 → certain. Impossible is $p = 0$ and certain is $p = 1$; everything else sits in between.

2. A distribution adds up to one · rewatch ▶ 0:26-0:43

2.1 (a) 1.00 (b) 0.60 (c) 0.20. $0.40 + 0.25 + 0.15 + 0.10 + 0.10 = 1.00$; $1 - 0.40 = 0.60$; $0.10 + 0.10 = 0.20$.

2.2 B. A adds up to 1.2 and D to only 0.9; C has a negative value. B is fine: 0 is allowed and the total is exactly 1.

2.3 False. The total would become 1.20. The other words must lose 0.20 between them.

2.4 0.05. $0.55 + 0.30 + 0.10 = 0.95$, and the whole distribution adds up to 1.

3. Sampling: a weighted spinner · rewatch ▶ 0:43-0:52

3.1 144° , 90° , 54° , 36° , 36° . Yes: 360° . Each angle is probability $\times 360^\circ$, and since the probabilities add up to 1, the angles add up to 360° .

3.2 0.2. $72 \div 360 = 0.2$: turn the rule around.

3.3 B. One spin is one random pick. Spins don't change the probabilities, and the next spin is a fresh random pick.

3.4 False. mat is picked most often (40 % of spins), but every word with a slice can come up.

4. Many spins match the probabilities · rewatch ▶ 0:53-1:02

4.1 mat 80, floor 50, sofa 30, bed 20, roof 20. Probability $\times 200$: $0.40 \times 200 = 80$, and so on. The five counts add up to 200.

4.2 About 2,470; 2,500 predicted. $0.247 \times 10,000 \approx 2,470$ (the exact count was 2,466); $0.25 \times 10,000 = 2,500$. Close, not exact.

4.3 mat 4, bed 3, sofa 2, floor 1, roof 0. bed: 30 % instead of 10 %. Not broken. Ten samples are far too few for the counts to settle; with 10,000 they match closely. (These are the ten samples from the video's code, concept 7.)

4.4 False. Close, but not exact: floor 24.7 % instead of 25 %, roof 9.8 % instead of 10 %.

5. Why sample? Variety · rewatch ► 1:02-1:10

5.1 0.60. Greedy always picks mat (0.40), so every other word counts: $1 - 0.40 = 0.60$.

5.2 About 3; with greedy, 0. $0.15 \times 20 = 3$. Greedy picks mat every single time, so sofa never appears.

5.3 B. Same prompt, same probabilities, but each run makes its own random picks.

5.4 True. Greedy has no randomness: it always takes the biggest probability.

6. Seeds make randomness repeatable · rewatch ► 1:10-1:22

6.1 False. Same seed, same recipe, same sequence, every time.

6.2 B. The seed doesn't change the distribution; it makes the random picks repeatable.

6.3 No; yes. Different seeds give different sequences (seed 8 starts with mat, not floor). Both still sample the same distribution, so over many samples both land near 40 %, 25 %, 15 %, 10 %, 10 %.

7. Sampling in NumPy · rewatch ► 1:23-1:43

7.1 B. Ten independent spins of the spinner.

7.2 True. Those add up to 1.1, not 1, so `choice` raises “ValueError: Probabilities do not sum to 1”.

7.3 (a) floor, then ['bed' 'sofa' 'mat' 'mat' 'bed' 'mat' 'bed' 'sofa' 'floor' 'mat'] · (b) ['floor' 'bed' 'sofa' 'mat' 'mat' 'bed' 'mat' 'bed' 'sofa' 'floor'] · (c) 0.3996, 0.2466, 0.1517, 0.1041, 0.098 (a) Yes: exactly the one word and the ten words shown in the video. (b) The same seed restarts the same sequence of random picks, so the ten words are the video's sequence from the start: *floor* followed by the first nine of the ten. Here it makes no difference whether you draw them one at a time or ten at once. (c) mat 0.3996, floor 0.2466, sofa 0.1517, bed 0.1041, roof 0.098: close to p , not exact. These are the video's 10,000-spin frequencies (40.0 %, 24.7 %, 15.2 %, 10.4 %, 9.8 %), which were simulated with this same seed.